MIL-C-14765(MU)

17 June 1968

SUPERSEDING
PA-PD-2296 Rev 1
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MILITARY SPECIFICATION

CABLE ASSEMBLIES SPECIAL PURPOSE ELECTRICAL T-4177, "NO GO"; T-4178, "GO"

1. SCOPE

1.1 This specification covers special requirements not included on the drawings, and provides the quality assurance provisions for the fabrication of parts, assembling, and the preparation for delivery for one type of cable assembly used to test XM-455 projectile and XM555 Fuze and designated as "NO GO" Cable Assembly Special Purpose T-4177 or "GO" Cable Assembly Special Purpose T-4178.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-S-60902(MU) - Soldering Electrical Connections for Nuclear Weapons Systems, Procedure for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1168 - Lot Numbering of Ammunition

F.S.C - 1190

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DRAWINGS

ORDNANCE CORPS

8794342 - Seal, Metallic, Assembly, Details and Instructions for Application
8800546 - Bag, Packing, Ammunition, for Cable Assembly
8806652 - Marking and Sealing for Cylindrical Containers
8857456 - Connector, Plug, Electrical
8860900 - Drum, Metal, Shipping (Interior)
8864700 - "GO" Cable Assembly Special Purpose Electrical: T-4178
8864701 - "NO GO" Cable Assembly, Special Purpose, Electrical: T-4177
8864768 - Cap, Dust
8865426 - Strain Relief
8865586 - Connector Plug, Electrical
8883520 - Drum, Metal, Shipping for Cable Assembly Special Purpose, Electrical: T-4177
8883521 - Drum, metal, shipping for Cable Assembly Special Purpose, Electrical: T-4178
ET-8864700 - Equipment Tabulation
ET-8864701 - Equipment Tabulation

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material used to fabricate any part complies with the specifications listed on the applicable drawing.

3.2 Assembly. The cable assemblies shall comply with all requirements specified on Drawing (dwgs.) 8864700 and 8864701 as applicable, with all requirements specified herein, and with the requirements of applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. The contractor may select and assign any number which complies with the drawing requirement. These numbers shall not be duplicated.

3.5 Mating assembly. The assemblies shall be capable of being completely mated and unmated with their proper receptacles, without the use of tools. This capability shall be verified as specified in 4.4.1.

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3.6 Circuit resistance. Circuit resistance between pins of the assemblies listed in Table I and II shall be 0.2 ohm max. when tested in accordance with 4.4.2. Current through the assemblies shall be limited to 100 milliamperes.

TABLE I

8864700 "Go" Training cable assembly

From Pin	To Pin
2	A
A	C
C	3
11	6
6	B
9	13
13	E

TABLE II

8864701 "No-Go" Training cable assembly

From Pin	To Pin
2	A
3	C
6	B
9	13
11	6
13	E

3.7 Insulation Resistance. The insulation resistance between each terminal and all other terminals and connector shells listed for the respective cable assemblies in Table III and IV shall be greater than 20 megohms at ambient and 10 megohms at high and low temperatures when tested as specified in 4.4.3. Cable Connector shells must be at ground potential (test fixture ground.)

TABLE III

8864700 "Go" Training cable assembly

Positive Potential Applies to Pin	All Other Pins & Connector Shell Minus Potential Except
A	2,3,C
B	6,11
E	9,13

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TABLE III (cont)

Positive Potential Applies to Pin	All Other Pins & Connector Shell Minus Potential Except
F	
D	
1	
4	
5	
7	
8	
10	
12	
14	

TABLE IV

8864701"No-Go" Training cable assembly

Positive Potential Applied to Pin	All Other Pins & Connector Shell Minus Potential Except
A	2
B	6, 11
C	3
E	9, 13
D	
F	
1	
4	
5	
7	
8	
10	
12	

3.8 Low temperature. The assemblies shall show no evidence of deterioration such as cracks in the insulation or other components when tested as specified in 4.4.4.for a minimum of 4 hours at-40 ± 5 degrees fahrenheit (^{o}F). While at this temperature the assembly shall be capable of withstanding a 90 ± 10 degrees bend and comply with 3.7. After completion of the test the assemblies shall comply with 3.5, 3.6 and 3.7 at a temperature of +80 ± 10°F without the use of forced heating.

3.9 High temperature. The assemblies shall withstand a temperature of +160 ± 5°F without evidence of deterioration, such as cracks or swelling, when tested as specified in 4.4.5. While at this temperature, the assemblies shall comply with 3.7. After completion of the test the assemblies shall comply with 3.5, 3.6 and 3.7 at +80 ± 10°F without the use of forced cooling.

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3.10 Vibration. The assemblies shall withstand vibration for 15 minutes in each of three mutually perpendicular axes without evidence of damage affecting function when tested as specified in 4.4.6. The frequencies and amplitudes shall conform with Table V. Following this test the assemblies shall comply with 3.5, 3.6 and 3.7.

TABLE V

Vibration frequencies, time cycle and amplitudes

Frequency Range (c.p.s. per cycle)	Time Cycle (minutes)	Constant Acceleration
20-80-20	15	.06 double amplitude

3.11 Pull test. Each cable termination shall withstand a pull of 15 pounds without evidence of physical damage such as loose insulation, cracking, breaking or stretching of molding or insulation, or any other change in physical characteristics when tested as specified in paragraph 4.4.7. Cable assemblies shall comply with 3.6 and 3.7 following this test.

3.12 Torque test. The cable termination (see dwg. 8857456) shall withstand a torque of 30 ± 5 inch-pounds (in-lb) without evidence of physical damage such as loose insulation, or cracking, breaking, or stretching of molding or insulation, or any other change in physical assemblies shall comply with 3.6 and 3.7 following this test.

3.13 Workmanship.

3.13.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.13.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.13.3 Burrs. Parts shall be free from burrs.

3.13.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.13.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

3.13.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise

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specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 3 complete assemblies (see dwgs. 8864700 and 8864701 as applicable) and 3 sets of parts, for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in table VI. Tests need not be conducted in order specified in table VI.

Table VI

First article inspections and tests

Examination or test	Contractor's inspection before submission	Non-Destructive	Parts	Assembly
1. Classification of characteristics (see 4.3.2.1)	X	X	X	X
2. Mating assembly (see 3.5)	X	X		X
3. Circuit resistance (see 3.6)	X	X		X
4. Insulation resistance (see 3.7)	X	X		X
5. Low temperature (see 3.8)	X	X		X
6. High temperature (see 3.9)	X	X		X
7. Vibration (see 3.10)	X	X		X
8. Pull test (see 3.11)	X	X		X
9. Torque test (see 3.12)	X	X		X

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4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Product acceptance inspection.

4.3.1 Lot formation. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.1 Inspection lot. An inspection lot shall be the quantity of parts, subassemblies, or assemblies from the same lot, which are submitted to inspection for Government acceptance.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table and inspected for all characteristics in the applicable classification. Any characteristic in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Major characteristic table VII

Lot size	Sample size
1- 25	100% inspection
26- 30	25
31- 35	26
36- 40	27
41- 45	28
46- 50	29
51- 55	30
56- 60	31
61- 65	32
66- 70	33
71- 75	34
76- 85	35
86- 95	36
96- 110	37
111- 130	38
131- 156	39
157- 190	40
191- 240	41
241- 340	42
341- 520	43
521-1120	44
1121 and over	45

Minor characteristic table VIII

Lot size	Sample size
1- 25	100% inspection
26- 35	16
36- 45	17
46- 55	18
56- 65	19
66-100	20
101-200	21
201 and over	22

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4.3.2.1 Classification of characteristics. Critical, major, and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Strain relief (see dwg. 8865426) covering a detail of dwgs. 8864700 and 8864701.

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Large outside diameter, min	Gage
102.	Wall thickness at small diameter, min	Gage
103.	Total length, max	Gage
104.	Length of thread, min	Gage
105.	Pitch diameter of thread, max	Gage
106.	Minor diameter of thread, max	Gage
107.	Three small holes present	Visual
108.	Length of large outside diameter, min	Gage
109.	Small inside diameter, max	Gage
110.	Material (see 3.1)	Visual
111.	Objective evidence (see 6.2) available for protective finish in accordance with dwg. 8865426	Visual
Minor:		
201.	Workmanship (see 3.13)	Visual/Manual

4.3.2.1.2 Connector, plug, electrical (see dwgs. 8857456 and 8865586), and dust cap (see dwg. 8864768) covering details of dwgs. 8864700 and 8864701.

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Objective evidence (see 6.2) available for part in accordance with applicable drawing	Visual
Minor:		
201.	Workmanship (see 3.13)	Visual/Manual

4.3.2.1.3 "Go" Cable assembly special purpose electrical T-4178 (see dwg. 8864700) and "No Go" cable assembly special purpose T-4177 (see dwg. 8864701) before assembly of strain relief, potting and molding.

Categories	Characteristics	Method of inspection
Critical: None defined.		

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Major:
101. Correctly wired.....Visual
102. Soldering in accordance with MIL-S-60902(MU).....Visual/Manual

Minor:
201. Workmanship (see 3.13).....Visual/Manual

4.3.2.1.4 "Go" Cable assembly special purpose electrical T-4178 (see dwg. 8864700) and "No Go" cable assembly special purpose T-4177 (see dwg. 8864701) before molding of connectors.

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. Cable tight in strain relief.....Manual
102. Connector tight in strain relief.....Manual
103. Proper potting.....Visual
104. Materials (see 3.1).....Visual

Minor: None defined.

4.3.2.1.5 "Go" Cable assembly special purpose electrical T-4178 (see dwg. 8864700) and "No Go" cable assembly special purpose T-4177 (see dwg. 8864701).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. Proper molding.....Visual/Manual
102. Total length.....*Measure
103. Dust cap present.....Visual
104. Materials (see 3.1).....Visual

*Measure with suitable scale.

Minor:
201. Location of marking.....Visual
202. Marking.....Visual
203. Serialization (see 3.4).....Visual
204. Workmanship (see 3.13).....Visual/Manual

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4.3.2.1.6 Bag, packing, ammunition, for cable assembly (see dwg. 8800546 covering a detail of dwgs. 883520 and 883521).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Length of envelope before assembly, min	*Measure
102.	Width of envelope before assembly, min	*Measure
103.	Envelope properly assembled and taped before assembly in bag	Visual/Manual
104.	Length of bag before assembly, min	*Measure
105.	Width of bag before assembly, min	*Measure
106.	Proper heat seal and assembly in bag	Visual/Manual
107.	Objective evidence (see 6.2) available for heat seal test in accordance with MIL-P-116	Visual
108.	No tear, split or crack in bag	Visual
109.	Materials (see 3.1)	Visual

*Measure with suitable scale.

Minor:		
201.	Marking	Visual
202.	Workmanship (see 3.13)	Visual/Manual

4.3.2.1.7 Drum, metal, shipping for cable assembly, special purpose, electrical: T-4177 (see dwg. 8883520) and drum, metal, shipping for cable assembly, special purpose, electrical: T-4178 (see dwg. 8883521) (see section 5).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Objective evidence (see 6.2) available for drum in accordance with dwg. 8860900	Visual
102.	Contents tight in drum	Manual
103.	Objective evidence (see 6.2) available for seal in accordance with dwg. 8794342	Visual
104.	Seal properly attached	Visual/Manual
105.	Cover assembled in accordance with dwg. 8806652	Visual/Manual
106.	Materials (see 3.1)	Visual
Minor:		
201.	Marking	Visual
202.	Workmanship (see 3.13)	Visual/Manual

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4.3.3 Testing. The contractor shall perform all the tests specified in this paragraph. Tests do not have to be performed in the order specified.

Test	Test classification
a. Mating Assembly	Major
b. Circuit Resistance	Major
c. Insulation Resistance	Major
d. Pull Test	Major
e. Torque Test	Major

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Mating Assembly (see 3.5). Circuit Resistance (see 3.6) and Insulation Resistance (see 3.7). Each assembly in the lot shall be subjected to these tests (see 4.4.1, 4.4.2, and 4.4.3 as applicable). If any assembly fails to comply with the applicable requirements, it shall be classified defective and removed from the lot.

4.3.3.2 Pull Test (see 3.11); Torque Test (see 3.12). Sample cable assemblies shall be selected from each inspection lot in accordance with Table IX and subjected to these tests as applicable (see paragraphs 4.4.7 and 4.4.8 as applicable). If any item in the sample fails to comply with the applicable requirements, the lot shall be rejected.

TABLE IX

Lot Size	Sample Size
1-18	100% inspection
19-30	18
31-50	27
51-70	32
71-90	34
91-110	36
111-130	37
131-150	38
151-180	39
181-220	40
221-290	41
291-430	42
431-820	43
821-2000	44
2001-and over	45

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL).

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EL's for assemblies are listed on the Equipment Tabulation, ET-8864700 or 8864701 as applicable. When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment. The contractor shall obtain approval of such designs, operating instructions, and special calibration instructions as specified in 6.3 prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired.
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL.

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on EL's. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Mating assembly (see 3.5). Each connector shall be assembled to its mating receptacle mounted rigidly on a fixture.

4.4.2 Circuit Resistance (see 3.6). The resistance between pins listed in 3.6 shall be measured with an approved electrical resistance tester.

4.4.3 Insulation Resistance (see 3.7). The insulation resistance between each terminal and each terminal and connector shell specified in 3.7, shall be measured with an approved test device which applies a potential of 500 $\pm$ 50 volts Direct Current, interlocked at 5 mils, for 15 seconds minimum or until the megohmeter or testing device stabilizes.

4.4.4 Low temperature (see 3.8). Place the assemblies in a chamber whose temperature is -40 $\pm$ 5°F for a minimum of 4 hours. Each cable shall be grasped using insulated gloves with approximately one half inch cable between the thumbs without removing the cable from the chamber. The cable shall be bent 90 $\pm$ 10 degrees and then straightened. After completion of the bending test the cable shall comply with requirement 3.7 without removing cable from the chamber. Remove the assembly from the chamber and allow the cable to return to +80 $\pm$ 10°F without using forced heating. The assemblies shall then comply with 3.5, 3.6, 3.7 and 3.8.

4.4.5 High temperature (see 3.9). The assemblies shall be placed in a chamber whose temperature is + 160 $\pm$ 5°F for a minimum of 4 hours. While at this temperature the assemblies shall comply with 3.7 without removing the assemblies from the chamber and allow them to return to +80 $\pm$ 10°F without the use of forced cooling. Test assemblies shall comply with 3.5, 3.6, and 3.7, and 3.9.

4.4.6 Vibration (see 3.10). The assemblies shall be mounted in a fixture attached to a suitable shake table by mounting the connectors rigidly to a vertical plane, the center of cable being three inches above horizontal plane of fixture. The cable shall be clamped to the horizontal plane nine inches from the vertical plane allowing the natural bend of the cable to take place. Using this fixture the cable shall be vibrated in three mutually perpendicular axes to the frequencies, cycles times, and amplitudes given in Table V, 3.10. After completion of test, the cable shall comply with 3.6, 3.7 and 3.10.

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4.4.7 Pull (see 3.11). The ends of the cable assembly shall be rigidly connected to mating receptacles or suitable fixtures which will transmit the pull through the metal portion of the connectors. A 15 ± 1 pound pull shall be applied and released by pulling the cable through its longitudinal axis six consecutive times so that the pull is transmitted through both connectors. After completion of this test, the cable assembly shall comply with 3.6, 3.7 and 3.11. The cable clamp shall not damage the cable when the pull is applied.

4.4.8 Torque test (see 3.12). The connector (see dwg. 8857456) end of the cable assembly shall be connected to a mating receptacle mounted on a suitable fixture. A torque of 30 ± 5 inch pounds shall be applied in a clockwise and then in a counterclockwise direction for a total of 3 times at a distance of 1/2 + 1/8 inch to the rear of the metal portion of the connector base. The torque applying device shall not damage the rubber molding on the cable termination. After completion of the test the cable assembly shall comply with 3.6, 3.7 and 3.12.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The cables shall be packaged in accordance with dwgs. 8883520 and 8883521 as applicable, using the specified moisture barrier.

5.1.2 Level C. The cables shall be packaged in accordance with dwgs. 8883520 and 8883521 as applicable, except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The cables shall be packed in accordance with dwg. 8883520 and 8883521 as applicable.

5.3 Marking. Marking shall be as specified on dwgs. 8883520 and 8883521 as applicable.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type required (see 1.2).
c. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment. The contractor shall obtain approval of equipment designs, operating instructions, and calibration instructions from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-NR4.

Custodian:
Army - MU

Preparing activity:
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